

# Organize the Freezer SOP (1x Freezer, in situ)

One episode organizes one freezer, in place.

The base is **passive**. It has no drive of its own. It is pushed by hand to the freezer and locked there. Nothing is carried away to a table. Everything the episode touches is already at the freezer when recording starts: the six items and the two bins inside it, and the work tray with the label sheet on the stand in front of the base.

The freezer is worked **as found**. The door stands open on its catch when recording starts. The working shelf carries two empty bins seated square against the back wall, with six items lying loose in a jumble in front of them. The episode ends with both bins seated square at the back wall again, each bin holding its own category stacked with the oldest item on top, two bags carrying a fresh label, the work tray empty, and the door pressed fully shut.

**This is an in-situ task. Three rules follow from that.**

First, there is a shelf directly above the working shelf. So **every approach to the working shelf is from the front, straight in, level**. No gripper comes down onto the working shelf from above. A gripper comes down from above only onto a bin that has been drawn forward to the shelf front edge, where there is clear space overhead.

Second, **the freezer and the door are never leaned on and never pushed**. No gripper, wrist, or forearm rests on the door, the shelf front edge, or a bin wall. No push is ever hard enough to move the appliance.

Third, **bins are slid, never lifted**. A bin is drawn forward and pushed home along the shelf board by its front rail. Neither bin ever leaves the working shelf.

**The order never changes: pull all six items out onto the work tray, relabel the two named bags, draw both bins forward, load each bin oldest up, slide both bins home, then close the door.** The items come out before the bins move, so nothing is carried over a bin. The relabeling happens while the bags still lie flat on the work tray, because a bag buried in a stack cannot be labeled.

**Every bin is loaded newest first and oldest last, so the oldest item ends on top.** In each bin the item marked **3** goes in first, then **2**, then **1** on top. This is what "oldest up" means everywhere in this document, and it is the same in both bins.

**Arm roles are fixed and never swap.** The **right gripper is the item gripper**: it pulls every item off the working shelf, lays each one on its tray spot, peels and presses both labels, lays every item into its bin, and presses the door home at the end. The **left gripper is the bin and door gripper**: it draws both bins forward, slides them home, holds the label sheet flat, holds a

bag flat during a label press, and pushes the door off its catch and round at the end. Neither arm reaches across the front of the other arm's body, and nothing is handed between grippers.

## Setup

Complete the base positioning and both checklists before starting an episode.

### Cell configuration

- **Environment camera:** 900 mm.
- **cell\_type:** bimanual

### Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the freezer. Stop it **square to the front of the appliance**, so the front edge of the working shelf runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the working shelf**, so the left end and the right end of the shelf are the same distance out from the middle of the base, and the two bins sit one either side of the middle.
3. Stop it **close enough** that both grippers reach the back wall of the working shelf straight in through the door mouth without an arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the appliance or the open door while both arms work.
4. Check the **height band**. With the base parked, both grippers enter the door mouth level and reach the working shelf board from front to back. None of this makes a wrist or forearm touch the shelf above.
5. Check the **right arm**. The **right gripper** reaches every one of the six items where they lie on the working shelf, both tray spots, the label sheet, and the door's front face at its free edge, without extending.
6. Check the **left arm**. The **left gripper** reaches both bin front rails at the back wall and again at the shelf front edge, the label sheet, the bag spot, and the door's inner face near the hinge, without extending and without reaching across the front of the right arm's body.
7. Check the **door catch**. The door stands open on its catch on its own, clear of the whole door mouth and clear of both arms' paths, and it does not drift while the arms work.
8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a freezer the arms cannot reach comfortably.

**The base stays locked and still for the whole episode.** Nothing moves it. No arm leans on the appliance hard enough to shift it. No one touches it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

## Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the door mouth. Its frame includes the whole working shelf, both bins seated and drawn forward, the shelf front edge, the open door, the door seal all round the mouth, and the work tray with the label sheet on its stand.
3. The camera reads the working shelf from the front. You can see which items are still inside, where each bin sits, and whether anything sticks out past the shelf front edge.
4. The camera reads each item's date number when the item lies on its tray spot.
5. The camera reads each loaded bin from above, so the order of the stack and the date number on the top item are both readable.
6. The camera reads the door front from the front once it is shut. You can see any gap along the seal.
7. Both arms are at home with grippers open.
8. The **right arm** reaches every item on the working shelf, both tray spots, the label sheet, and the door's front face at its free edge, without extending to a joint limit. It never needs to reach a bin front rail, the door catch, or the door's inner face.
9. The **left arm** reaches both bin front rails at the back wall and again at the shelf front edge, the label sheet, the bag spot, and the door's inner face near the hinge, without extending to a joint limit. It never needs to reach an item on the working shelf or the box spot.
10. Each gripper enters and leaves the door mouth **from the front and level**. Neither gripper comes down onto the working shelf from above, and neither wrist nor forearm touches the shelf above on the way in or out.
11. A gripper comes straight down from above only onto a bin standing at the shelf front edge, where there is clear space over the bin rim.
12. The two arms do not touch each other, and neither reaches across the front of the other arm's body. The open door on its catch is clear of both arms' paths to the door mouth and to the work tray.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 12 hold.

## Materials checklist

1. The **freezer** stands where it lives. It is not moved, not leaned on, and not pushed at any point.
2. The **door** is hinged on the **left** of the appliance. At the start of the episode it stands open to the left, held by a **door catch**, wide enough that the whole door mouth is clear.

3. The **door catch** holds the door open on its own. The door does not swing in or drift out while the arms work, and it comes off the catch under one steady push without being unhooked by hand.
4. The door shuts with a latch or a magnet seal that holds it closed. A correctly shut door meets the cabinet all round, shows no gap along the seal, and does not spring back open.
5. The **working shelf** is the lowest shelf of the freezer, with a **shelf directly above it**. So the only way in is the front-facing door mouth, never the top.
6. The working shelf is one flat board. There is no drawer and nothing else on it beyond the two bins and the six items.
7. The working shelf is deep enough front to back for a bin to stand against the back wall and to stand again at the shelf front edge, and wide enough for the two bins to stand side by side with clear space between them.
8. There is clear space above both bins, so a loaded bin stands on the shelf without its stack touching the shelf above.
9. **Two bins** stand on the working shelf, side by side, seated square against the back wall, with clear space between them. Both start empty.
10. The **bag bin** is the left one and carries a blue **BAGS** label on its front rail. The **box bin** is the right one and carries a green **BOXES** label on its front rail.
11. Each bin is open topped, rigid, and light. Each has a **front rail** — a raised top edge along its front wall that a gripper closes on. A bin slides on the shelf board when drawn by its front rail and does not need to be lifted.
12. Each bin is deep enough to hold three items stacked flat with the top item sitting below the bin rim, and is no wider than one item, so a stack has one place to go and no choice of lane.
13. **Six items** lie flat on the working shelf in front of the bins, in one row across the board, none touching another, none on top of another, each with its date number facing up. They lie in this fixed order, left to right: **box 2, bag 3, box 1, bag 1, box 3, bag 2**.
14. No item sticks out past the shelf front edge at the start.
15. **Three items are bags** and **three items are boxes**. A bag is a flat sealed freezer bag that keeps its shape when gripped and lies flat on one face. A box is a rigid carton that lies flat on one face.
16. Every item carries a **date tag** on the face that starts upward, printed with one **date number**: 1, 2, or 3. Each category uses each number once, so there is a bag 1, a bag 2, a bag 3, a box 1, a box 2, and a box 3.
17. Every item is closed, dry, and sealed, and is held by one gripper without its contents spilling or its seal opening.
18. Every bag carries a blank **label patch** on the same face as its date tag, beside the date tag and not overlapping it.

19. The **work stand** stands on the floor directly in front of the base, below and clear of the door mouth, at a height where both grippers work on it without extending. It does not slide when an item is laid on it or lifted off it.
20. The **work tray** lies flat on the stand, its face up, holding the **bag spot** at its center and the **box spot** at its right, side by side and not touching. It is empty at the start.
21. Each tray spot is large enough for three items to lie flat side by side with clear space between them.
22. One **label sheet** lies flat on the work tray at its left, to the left of the bag spot, written side up, with no corner lifted.
23. The label sheet carries **two labels** side by side. The **left label names bag 2** and the **right label names bag 3**. Each label is written large enough for the camera to read the bag it names.
24. The floor and the space between the stand and the door mouth are clear, so an arm carrying an item passes over nothing.

## Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the appliance itself: the door mouth, the shelf front edge, the back wall, and the bins standing on the shelf.

- **Freezer:** the appliance the base is parked at. It is never moved, never leaned on, and never pushed.
- **Door mouth:** the open front of the freezer. Both grippers go in and come out through it, level and from the front.
- **Working shelf:** the lowest shelf, and the only shelf worked. All shelf work happens on it.
- **Shelf above:** the shelf directly above the working shelf. It is what makes every shelf approach a front approach. Nothing ever touches it.
- **Shelf front edge:** the front edge of the working shelf board. Nothing may stick out past it, or the door will not shut.
- **Back wall:** the inside back wall of the freezer. Both bins are seated square against it at the start and again at the end.
- **Bag bin:** the left bin on the working shelf, labeled **BAGS** in blue. It takes the three bags. **Left gripper moves it; right gripper loads it.**
- **Box bin:** the right bin on the working shelf, labeled **BOXES** in green. It takes the three boxes. **Left gripper moves it; right gripper loads it.**
- **Work stand:** the stand in front of the base holding the work tray. It is never moved.
- **Work tray:** the flat tray on the stand, holding the two tray spots and the label sheet. Empty of items at the start and empty again at the end.

- **Bag spot:** the center of the work tray. Every bag lands here after it is pulled. **Right gripper lays items here; left gripper only holds a bag flat during a label press.**
- **Box spot:** the right of the work tray. Every box lands here after it is pulled. **Right gripper only.**
- **Label sheet:** flat on the work tray at its left, written side up. It stays flat on the tray for the whole episode.
- **Door:** the freezer door, hinged on the left, held open on its **door catch** for the whole of the work. Neither gripper touches it until Step 6.
- **Door catch:** the stay that holds the door open on its own. The door comes off it under one steady push in Step 6.1.

## Arm assignments

- **Right gripper — the item gripper.** Pulls each of the six items off the working shelf, lays each one on its tray spot, peels each label off the label sheet and presses it onto its bag, lays each item into its bin, and presses the door home at the end. It never touches a bin, the door catch, or the door's inner face.
- **Left gripper — the bin and door gripper.** Draws each bin forward to the shelf front edge, pushes each bin home to the back wall, holds the label sheet flat during a peel, holds a bag flat during a label press, and pushes the door off its catch and round to the door mouth at the end. It never carries an item and never lays an item into a bin.
- Neither arm reaches over, under, around, or past the other, and neither reaches across the front of the other arm's body. Nothing is handed between grippers, and only one arm moves an object at a time.

## Vocabulary

- **Date number:** the number printed on an item's date tag — 1, 2, or 3. **1 is the oldest and 3 is the newest.** There is one of each number in each category.
- **Oldest up:** the item marked 1 ends on top of that bin's stack. It is reached by laying 3 in first, then 2 on top of it, then 1 on top of that. The same order is used in both bins.
- **Stack:** the items lying flat one on top of another in a bin, all facing up, all lined up with the bin walls.
- **Number up:** the item lies on the face opposite its date tag, so the date number faces the ceiling and the camera can read it.
- **Lies flat:** the item rests on one whole face, not on an edge, not leaning, and not propped on another item.
- **Front approach:** the gripper enters and leaves the door mouth level and from the front. It never comes down onto the working shelf from above. Every action on the working shelf is a front approach.

- **Front rail:** the raised top edge along a bin's front wall. It is the only part of a bin a gripper closes on.
- **Seated:** the bin stands on the working shelf with its back against the back wall and its walls lined up with the freezer walls, not turned and not tilted.
- **Square:** lined up with the walls it sits against, not turned to one side.
- **Drawn forward:** the bin has been slid straight toward the front along the working shelf until its front rail stands just inside the shelf front edge, still lined up square.
- **Stands proud:** part of an item sits above the bin rim, or part of a bin or an item sticks out past the shelf front edge.
- **Clear space:** two things do not touch each other, and the gap between them is visible from the front.
- **Label patch:** the blank rectangle printed on a bag's face beside its date tag. A new label goes here and nowhere else.
- **Label face:** the written side of a label, naming the bag it belongs to by that bag's date number.
- **Steady hold:** the left gripper closed and pressed down on a bag lying flat on the work tray, held still for the whole of the label press so the bag does not slide.
- **Sealed:** the door meets the cabinet all round, no gap shows anywhere along the seal, the latch holds, and the door does not spring back when the **right gripper** comes off.

## Steps

### Step 1: Pull the six items onto the work tray

**Goal:** the working shelf holds nothing but the two seated bins, and all six items lie flat on the work tray, bags on the bag spot and boxes on the box spot, every date number facing up.

Work the items from the left end of the row to the right end, one at a time. Do not skip one and come back to it. The **left arm stays at home** for the whole of this step. The door stands open on its catch and nothing touches it.

#### 1.1 Take one item off the working shelf

- With the **right gripper**, come in straight from the front through the door mouth, level, up to the next item in the row. Do not come down onto the shelf from above.
- Close on the **body** of the item, across its middle, not on a corner and not on its seal.
- Lift the item just clear of the shelf board, then carry it straight out to the front through the door mouth, level and low.
- Do not pass over a bin. Do not let the item touch another item, the shelf above, or the door on the way out.

**Check:** the item is out through the door mouth, the items still on the shelf lie where they were, and both bins are still seated square. If an item was dragged out of the row, put it back flat in the row with the **right gripper** before going on.

## 1.2 Lay the item on its tray spot

- Carry the item level down to the work tray.
- Lay a **bag** flat on the **bag spot** and a **box** flat on the **box spot**, number up, with clear space from anything already lying there.
- Set the item down on the tray before opening the **right gripper**. Never drop it.
- Draw the **right gripper** straight up clear of the tray.

**Check:** the item lies flat on the right spot, its date number faces up, and it touches nothing else on the tray. If it landed number down, pick it up with the **right gripper**, turn it over, and lay it flat again. If it landed on the wrong spot, pick it up and lay it on the right one.

**Expected state after six passes:** the working shelf holds the two seated bins and nothing else. Three bags lie on the bag spot and three boxes lie on the box spot, all flat and all number up. The door still stands open on its catch.

## Step 2: Relabel the two named bags

**Goal:** bag 2 and bag 3 each carry the label that names them, stuck flat on the label patch with no lifted corner.

Both labels are worked on the tray, where the bags lie flat. Do the **left label first, then the right label**, in that order.

### 2.1 Peel one label

- With the **left gripper** closed, press down on the label sheet beside the label being peeled and hold it flat on the tray. Hold it there until the label is off.
- With the **right gripper**, close on the free corner of that label.
- Peel the label off the sheet in one steady pull, low and flat across the sheet. Do not lift it straight up. Do not let it fold onto itself.
- Lift the peeled label clear of the sheet, holding it by the corner, label face up.

**Check:** the label is off the sheet in one piece, the sheet still lies flat on the tray, and the label face is readable. If the label folds onto itself, do not use it. End the episode and report it.

### 2.2 Press the label onto its bag

- Read the label face. It names bag 2 or bag 3.
- With the **left gripper** closed, take a **steady hold** on that bag where it lies flat on the bag spot, pressing down clear of its label patch.



- With the **right gripper**, lay the label flat onto the bag's **label patch**, square with the patch, then press it down.
- Press for 2 seconds, straight down. Then run the closed **right gripper** once across the label from one edge to the other, still pressing.
- Open the **right gripper** and lift it clear. Then lift the **left gripper** off the bag.

**Check:** the label sits on the label patch, square, flat, and with no corner lifted, and the bag has not slid on the tray. If a corner stands up, press it down again with the **right gripper** while the **left gripper** holds the bag.

### 2.3 Repeat for the second label

- Run Steps 2.1 and 2.2 again for the right label on the sheet, onto the bag it names.

**Check:** both labels are off the sheet and stuck flat, one on bag 2 and one on bag 3. The sheet lies flat and empty on the tray. Bag 1 carries no new label.

**Expected state:** six items on the tray, two of the bags freshly labeled, the label sheet flat and empty, both bins still seated square at the back wall.

### Step 3: Draw both bins forward

**Goal:** both bins stand square at the shelf front edge with clear space over each rim, ready to be loaded from above.

The **left gripper** does this whole step. The **right arm stays at home**. Work the **bag bin first, then the box bin**.

- With the **left gripper**, come in straight from the front through the door mouth, level, and close on the bin's **front rail**.
- Draw the bin straight toward the front along the shelf board until its front rail stands just inside the **shelf front edge**. Keep it square the whole way.
- Do not lift the bin. Do not turn it. Do not tip it. Do not let it stand proud of the shelf front edge.
- Open the **left gripper** and draw it straight out through the door mouth.
- Do the same with the box bin.

**Check:** both bins stand square at the shelf front edge, neither sticking out past that edge, with clear space between them and clear space over each rim. If a bin came round out of square, square it with the closed **left gripper** against its walls. If a bin stands proud of the shelf front edge, push it back a little by its front rail.

**Expected state:** two empty bins standing square at the shelf front edge, six items on the tray, the door still open on its catch.

## Step 4: Load each bin oldest up

**Goal:** each bin holds its three items stacked flat and square, in the order 3 at the bottom, then 2, then 1 on top, with the top item below the bin rim.

The **right gripper** does all the loading. The **left arm stays at home** for the whole of this step. Load the **bag bin first, then the box bin**. Load one item at a time, in the fixed order **3, then 2, then 1**.

### 4.1 Load the bag bin

- With the **right gripper**, close on the body of **bag 3** where it lies on the bag spot, across its middle.
- Lift it straight up, carry it level to the door mouth, and take it in from the front.
- Lower it straight down into the **bag bin** and lay it flat on the bin floor, number up, lined up with the bin walls.
- Set it down on the bin floor before opening the **right gripper**. Never drop or throw an item into a bin.
- Draw the **right gripper** straight up clear of the bin rim, then straight out through the door mouth.
- Do the same with **bag 2** onto bag 3, then **bag 1** onto bag 2.
- Do not carry an item over the box bin, over the door, or over the shelf front edge with an item hanging low.

**Check:** three bags lie flat in the bag bin, squared with the bin walls, with **1 on top** and its number readable from above. Nothing stands proud of the bin rim. If the stack is out of order, take the items out with the **right gripper**, lay them back flat on the bag spot, and load them again in the order 3, 2, 1.

### 4.2 Load the box bin

- Run the same motions as Step 4.1 with the **right gripper**, using **box 3**, then **box 2**, then **box 1** into the **box bin**.
- Do not carry a box over the loaded bag bin.

**Check:** three boxes lie flat in the box bin, squared with the bin walls, with **1 on top** and its number readable from above. Nothing stands proud of the bin rim. The work tray holds nothing but the empty label sheet.

**Expected state:** both bins loaded and standing square at the shelf front edge, oldest item on top in each, the tray empty of items.

## Step 5: Reseat both bins

**Goal:** both bins are seated square against the back wall with clear space between them, and nothing sticks out past the shelf front edge.

The **left gripper** does this whole step. The **right arm stays at home**. Work the **box bin first, then the bag bin**, so the **left gripper** never reaches past a bin it has just pushed home.

- With the **left gripper**, come in straight from the front through the door mouth, level, and close on the bin's **front rail**.
- Push the bin straight back along the shelf board until its back meets the **back wall**. Keep it square the whole way.
- Push until the bin stops. Do not push hard enough to move the appliance. Do not lift the bin and do not lean on its rim.
- Open the **left gripper** and draw it straight out through the door mouth.
- Do the same with the bag bin.

**Check:** both bins are **seated** square against the back wall, with clear space between them, nothing standing proud of the shelf front edge, no item standing proud of a bin rim, and the stacks still in order with 1 on top in each. The freezer has not moved. If a bin sits away from the back wall or came round out of square, draw it forward a little by its front rail with the **left gripper** and push it home again.

**Expected state:** two loaded bins seated square at the back wall, the working shelf clear of loose items, the tray empty of items, the door still open on its catch.

## **Step 6: Close the door fully**

**Goal:** the door is shut and sealed, with no gap along the seal and no spring back.

The door swings from the left, so the **left gripper** pushes it round. It seals at its free edge on the right, so the **right gripper** presses it home.

### **6.1 Push the door round**

- Check first that both grippers are out of the door mouth and empty, that both bins are seated square, and that nothing sticks out past the shelf front edge.
- With the **left gripper** closed, set it on the door's **inner face low down near the hinge**.
- Push steadily until the door comes off its **door catch**, then keep pushing it round toward the door mouth in one smooth move. Do not push fast. Do not let the door hit the appliance.
- Stop when the door has come round in front of the door mouth and rests against the cabinet.
- Draw the **left gripper** straight back off the door and keep it clear until the episode ends.

**Check:** the door hangs in front of the door mouth, nothing is caught between the door and the appliance, and nothing inside has been knocked over. If a stack was knocked over, push the door open onto its catch again with the **left gripper**, put the stack right with the **right gripper** in the order 3, 2, 1, and push the door round once more.

## 6.2 Press the door home

- With the **right gripper** closed, press the door's front face flat, at the **free edge side on the right**, straight back toward the appliance.
- Press until the door meets the cabinet all round and the latch holds. Hold the press for 2 seconds.
- Press straight back only. Do not slam the door. Do not press at an angle or on the door edge. Do not press hard enough to move the appliance.
- Draw the **right gripper** straight back off the door.

**Check:** the door is **sealed**. It meets the cabinet all round, no gap shows anywhere along the seal, and the door does not spring back when the **right gripper** comes off.

If a gap shows or the door springs back, press it home once more with the **right gripper**. If it still will not seal, push the door open onto its catch with the **left gripper** and look for a bin standing proud of the shelf front edge or an item standing proud of a bin rim.

**Expected state:** door shut and sealed. Both bins seated at the back wall. Work tray empty of items with the label sheet lying flat on it. Base still locked in its park.

## Step 7: End the episode

- Confirm the door. It is shut and sealed, with no gap along the seal and no spring back.
- Confirm the work tray. It holds no items, and the label sheet lies flat and empty where it started.
- Confirm nothing has been left on the stand or dropped on the floor.
- Confirm the base has not moved and the appliance has not shifted.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

## After the episode: reset the workspace

This reset is not recorded.

1. Open the door by hand and set it on its catch, wide enough that the whole door mouth is clear.
2. Draw both bins forward by hand and take the six items out.
3. Peel the two used labels off bag 2 and bag 3, and clean each label patch so it is blank and dry again.
4. Push both empty bins back to the back wall, seated square and side by side, with clear space between them.
5. Lay the six items flat on the working shelf in front of the bins, in one row, number up, none touching and none on top of another, in this order left to right: **box 2, bag 3, box 1, bag 1,**

### **box 3, bag 2.**

6. Check no item sticks out past the shelf front edge.
7. Put a fresh label sheet flat on the work tray at its left, written side up, carrying two labels that name bag 2 and bag 3.
8. Wipe up any spill or frost on the working shelf, in a bin, or on the tray, and dry it.
9. Replace an item that leaks, whose seal has opened, that will no longer lie flat on one face, or whose date number can no longer be read. Replace a bag whose label patch will no longer take a label.
10. Check the door still latches and seals when it is pushed shut by hand, that it does not spring back, and that its catch still holds it open on its own.
11. Check both bins still slide on the shelf board when drawn by the front rail, and that neither tips.
12. Check the work stand is still steady and does not slide when an item is laid on it or lifted off it.
13. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

## **SOP violations**

These are actions that break the SOP and are reviewed side by side in the review tool.

### **How to record a violation in review**

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

### **Episode handling**

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

## **Violations**

### **Violation: Base moved during the episode**

- **Visible cue:** the freezer shifts in frame, the door mouth changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

### **Violation: Approached the shelf from above or touched the shelf above**

- **Visible cue:** a gripper comes down onto the working shelf from above instead of entering level from the front. Or a wrist or forearm knocks, scrapes, or rests on the shelf above or on the shelf front edge on the way in or out.
- **SOP rule broken:** Steps 1.1, 3, and 5. Every gripper enters and leaves the door mouth level and from the front, and comes down from above only onto a bin standing at the shelf front edge.
- **Coaching note:** go in level and come out level. The only thing you reach down into is a bin standing at the front edge.

#### **Violation: Leaned or pushed on the freezer**

- **Visible cue:** a gripper, wrist, or forearm rests on the door, the shelf front edge, a bin wall, or the cabinet. Or a push on a bin or the door is hard enough to shift the appliance in frame.
- **SOP rule broken:** Steps 1 to 6. The freezer is never leaned on and never pushed, and no push moves the appliance.
- **Coaching note:** the appliance carries nothing but itself. Push a bin to its stop and stop pushing.

#### **Violation: Wrong work order**

- **Visible cue:** a bin is moved before all six items are out. A label is peeled or pressed after an item has gone into a bin. A bin is loaded before both bins are drawn forward. A bin is reseated before both bins are loaded. The door is pushed round before both bins are seated.
- **SOP rule broken:** Steps 1 to 6. Pull all six items, relabel the two bags, draw both bins forward, load both bins, reseal both bins, then close the door.
- **Coaching note:** items out first, labels next, bins last. A bag under a stack cannot be labeled.

#### **Violation: Items left on the working shelf**

- **Visible cue:** a bin is drawn forward while an item is still lying loose on the working shelf, or the episode ends with an item still on the shelf board.
- **SOP rule broken:** Step 1. All six items come off the working shelf onto the work tray before any bin is moved.
- **Coaching note:** count six on the tray before you touch a bin.

#### **Violation: Item laid on the wrong tray spot**

- **Visible cue:** a bag lies on the box spot or a box lies on the bag spot, or an item lies on the label sheet or off both spots on the tray.
- **SOP rule broken:** Step 1.2. Bags go flat on the bag spot and boxes go flat on the box spot.
- **Coaching note:** read the item before you set it down. Bags center, boxes right.

### **Violation: Item not laid flat or number not up**

- **Visible cue:** an item on a tray spot or in a bin lies on its edge, leans, sits propped on another item, or lies with its date number face down.
- **SOP rule broken:** Steps 1.2 and 4. Every item lies flat on one whole face with its date number up.
- **Coaching note:** flat and number up every time. If the number is not readable, the camera cannot score the stack.

### **Violation: Item gripped by a corner or a seal**

- **Visible cue:** the **right gripper** closes on an item's corner, its seal, or its date tag instead of across the middle of its body.
- **SOP rule broken:** Steps 1.1 and 4. The **right gripper** closes on the body of the item, across its middle, not on a corner and not on its seal.
- **Coaching note:** grip the middle of the body. A corner grip tips the item and a seal grip opens it.

### **Violation: Label sheet not held down**

- **Visible cue:** the **left gripper** is not pressing the label sheet flat while a label is peeled, the sheet lifts, slides, or comes off the tray during the peel, or the label is pulled straight up instead of low and flat.
- **SOP rule broken:** Step 2.1. The **left gripper** holds the sheet flat on the tray until the label is off, and the peel is one steady pull, low and flat.
- **Coaching note:** hold the sheet first, then peel low and flat. A sheet that moves folds the label.

### **Violation: Label put on the wrong bag or the wrong place**

- **Visible cue:** a label is pressed onto bag 1, onto a box, onto a bag it does not name, or anywhere on a bag other than its blank label patch — over the date tag, over the seal, or across an edge.
- **SOP rule broken:** Step 2.2. Read the label face, then press it square onto the label patch of the bag it names.
- **Coaching note:** read the label, find the patch, then press. There is one right patch for each label.

### **Violation: Label press done wrong**

- **Visible cue:** the **left gripper** does not hold the bag during the press, the bag slides on the tray, the press is shorter than 2 seconds, the closed **right gripper** is not run across the label, or the label is left with a lifted corner or a fold.

- **SOP rule broken:** Step 2.2. Hold the bag with the **left gripper**, press straight down for 2 seconds, then run the closed **right gripper** across the label.
- **Coaching note:** hold, press two seconds, sweep across, then look for a lifted corner.

**Violation: Bin lifted, turned, or not drawn to the shelf front edge**

- **Visible cue:** the **left gripper** lifts a bin off the shelf board, closes on a bin wall or rim instead of its front rail, or tips it. Or a bin is loaded while still short of the shelf front edge, or is left standing proud of that edge or out of square.
- **SOP rule broken:** Step 3. The **left gripper** draws each bin by its **front rail**, straight and square, along the shelf board until its rail stands just inside the shelf front edge.
- **Coaching note:** rail only, slide only. A bin that is lifted comes back down out of square.

**Violation: Item put in the wrong bin**

- **Visible cue:** a bag is laid into the box bin, or a box is laid into the bag bin.
- **SOP rule broken:** Step 4. Bags go into the **bag bin** and boxes go into the **box bin**.
- **Coaching note:** read the bin rail before the item goes down. BAGS is blue on the left, BOXES is green on the right.

**Violation: Oldest not ended on top**

- **Visible cue:** a bin is loaded in an order other than 3, then 2, then 1, or a finished stack shows a number other than **1** on top when read from above.
- **SOP rule broken:** Step 4. Load 3 first, then 2, then 1, so the oldest item ends on top in both bins.
- **Coaching note:** three, two, one, every bin, every time. Read the top number before you move on.

**Violation: Stack not square or standing proud of the bin rim**

- **Visible cue:** items in a bin sit turned, crossed, overhanging a bin wall, or stacked high enough that the top item stands above the bin rim.
- **SOP rule broken:** Step 4. Each item is laid flat and lined up with the bin walls, and the top item sits below the bin rim.
- **Coaching note:** line each item up with the walls as it goes down. A crossed item lifts the whole stack.

**Violation: Item dropped into a bin**

- **Visible cue:** the **right gripper** opens above the bin floor or above the item below, and the item falls the last part of the way.
- **SOP rule broken:** Step 4. Set the item down on the bin floor or on the item below before opening the **right gripper**.



- **Coaching note:** take it all the way down, then open. A dropped item lands out of square.

#### **Violation: Item carried over a bin or the door, or more than one moved at once**

- **Visible cue:** an item is carried over a bin, over the door, or over another item on the working shelf. Or two items are held or moved at once, or one item is dragged along the shelf board or the tray instead of being carried.
- **SOP rule broken:** Steps 1 and 4. Carry one item at a time, level and low, straight in and out through the door mouth, and never over a bin, the door, or another item.
- **Coaching note:** one item, one clear path. Anything carried over a bin lands in it when it slips.

#### **Violation: Bin not reseated home and square**

- **Visible cue:** the episode ends with a bin short of the back wall, turned out of square, tilted, sticking out past the shelf front edge, or touching the other bin.
- **SOP rule broken:** Step 5. The **left gripper** pushes each bin by its front rail straight back until it is seated square against the back wall, with clear space between the bins.
- **Coaching note:** push it to the wall and check the gap. A bin left short holds the door open.

#### **Violation: Door touched or moved during the work**

- **Visible cue:** a gripper takes hold of the door, leans on it, or pushes it during Steps 1 to 5. The door comes off its catch before Step 6. Or the door drifts in or out while the shelf, the bins, or the tray are worked.
- **SOP rule broken:** Steps 1 to 5. The **door catch** holds the door open, and neither gripper touches the door until Step 6.
- **Coaching note:** the catch holds the door. Both arms are for items and bins until the work is done.

#### **Violation: Door pushed round early or into a blocked mouth**

- **Visible cue:** the **left gripper** pushes the door off its catch while an item is still held, while an arm is still inside the door mouth, while a bin is still short of the back wall, or while something sticks out past the shelf front edge. Or the door is pushed fast, is let swing on its own, or hits the appliance.
- **SOP rule broken:** Step 6.1. Check the mouth is clear and both bins are seated, then push the door round steadily with the **left gripper** on its inner face near the hinge.
- **Coaching note:** clear the mouth, then move the door slowly. A fast door knocks a stack over.

#### **Violation: Door not closed and sealed**

- **Visible cue:** the episode ends with the door open or ajar, a gap shows along the seal, the latch has not held, or the door springs back when the **right gripper** comes off.

- **SOP rule broken:** Step 6.2. The **right gripper** presses the door's front face flat and straight back at the free edge, holds the press for 2 seconds, and the door meets the cabinet all round with the latch holding.
- **Coaching note:** flat face, straight back, hold two seconds, then watch it before you move away. A door left ajar is a failed episode.

#### **Violation: Required check not followed**

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — an item on the wrong spot, an item number down, a label with a lifted corner, an item in the wrong bin, a stack out of order, a bin short of the back wall, or a door showing a gap.
- **SOP rule broken:** Steps 1, 2, 3, 4, 5, 6.1, and 6.2. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

#### **Violation: Dropped, toppled, or knocked over**

- **Visible cue:** an item is dropped on the tray, on the shelf, or on the floor. An item already laid down is knocked off its tray spot or out of a bin. A stack is toppled by an arm or by the door. The label sheet is struck and moved off the tray. Or a bin is knocked out of square.
- **SOP rule broken:** Steps 1 to 6. Nothing is dropped, toppled, or knocked out of its place, and every gripper leaves the door mouth straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

#### **Violation: Wrong arm used**

- **Visible cue:** the **right gripper** touches a bin, the door catch, or the door's inner face. The **left gripper** picks up or carries an item, lays an item into a bin, or touches the box spot. An item is handed from one gripper to the other. Or one arm reaches across the front of the other arm's body.
- **SOP rule broken:** Steps 1 to 6. The **right gripper** works the items, the labels, and the door press. The **left gripper** works the bins, the label sheet, the bag hold, and the door push. Nothing changes hands.
- **Coaching note:** right arm on items, left arm on bins and the door. Nothing crosses and nothing is passed.

#### **Violation: Wrong episode ending**

- **Visible cue:** the episode ends with the door open or showing a gap, an item still on a tray spot or on the working shelf, a label still on the sheet, a bin short of the back wall or standing proud of the shelf front edge, a stack with a number other than 1 on top, an arm short of home, or a gripper not fully open.

- **SOP rule broken:** Step 7. Confirm the door, the tray, the base, and the floor, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

## Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read a date number, a bin rail label, a label face, or the door seal**, so which item went where or whether the door shut cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Faulty door:** a door whose latch will not hold, whose seal will not close under a correct press, or that swings shut on its own off the door catch.
- **Door catch will not hold:** a catch that lets the door drift in or out while the arms work.
- **Freezer will not stay put:** an appliance that shifts under a correct bin push or a correct door press.
- **Faulty bin:** a bin that will not slide on the shelf board, that tips when drawn by its front rail, that will not sit square against the back wall, or whose front rail breaks under a correct grip.
- **Defective item:** one that leaks, whose seal opens in a correct grip, that will not lie flat on one face, that slips out of a correct grip, or whose date number cannot be read.
- **Faulty label or label sheet:** a label that tears or folds onto itself in a correct peel, a label that will not stick to a clean label patch, a sheet whose two labels do not name bag 2 and bag 3, or a sheet that has fewer or more than two labels.
- **Bin too small:** a bin that will not hold its three items stacked flat with the top item below the bin rim.
- **Working shelf too small:** a shelf that will not take both bins side by side with clear space between them, or that will not let a bin stand at the shelf front edge and again at the back wall.
- **Work stand will not stay put:** a stand that slides when an item is laid on it or lifted off it.
- **A place turns out to sit outside its arm's comfortable reach**, so the working shelf, a bin at either position, the label sheet, a tray spot, or the door cannot be reached without extending the arm.

## Annotation subtasks (from SOP)

1. Take one item off the working shelf
2. Lay one item flat on its tray spot

3. Hold the label sheet flat and peel one label
4. Press one label onto its bag's label patch
5. Draw one bin forward to the shelf front edge
6. Lay one item flat into its bin
7. Build one bin stack in the order 3, 2, 1
8. Push one bin home to the back wall
9. Push the door off its catch and round to the door mouth
10. Press the door home until it seals
11. Return both arms home and end the episode